

Biology

M.C.Q.s

1. C	Q. 2 (i)	
2. C	A. 2 (i)	
3. A	Taxa	Pea
4. B	Kingdom	Plantae
5. C		
6. A	Division	Magnoliophyta
7. D		
8. B	Class	Magnopsida
9. A		
10. C	Order	Fabales
11. D		
12. B	family	Fabaceae
13. C		
14. B	Genus	Pisum
15. D		
16. B	Species	Pisum sativum
17. A		
18. C		
19. B		

20. B	Q. 2 (ii)	
21. A	A. 2 (ii)	
22. C	Advantages:	
23. B	(1) Flexibility: The cell	
24. D	can change shape easily	
	(2) Movement: cell can	
	move or change position	
	more easily	

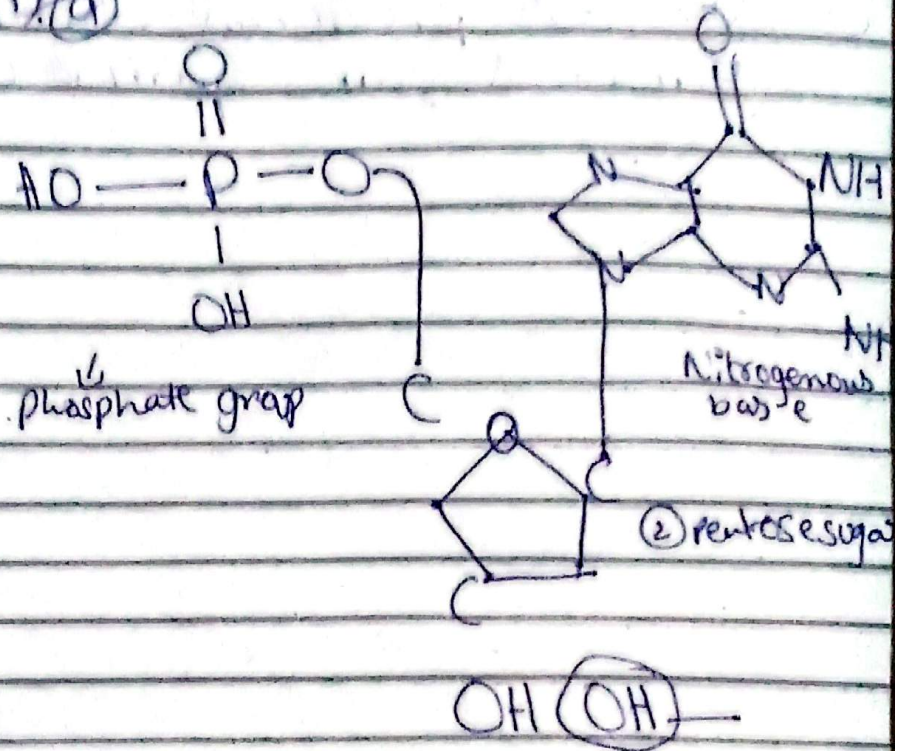
Endocytosis... cell can engulf substances, such as food particles, through its flexible cell membrane.

Q: 2(iii)

A: 2(ii)

Biomolecules	Location in cell	Roles
Carbohydrate eg Glucose, lactose, maltose & Sucrose	cell wall, cell membrane, SER, golgi apparatus	Used as food, energy source
Proteins e.g. haemoglobin, Actin, myosin	Cytoplasm	Energy source
Lipids e.g. phospholipids	cell membrane, SER	provides cell with long-term energy
DNA	Nucleus (chromosomes)	Tools for translation & protein synthesis.

Q: 3. (i) (a)



* Phosphate group attached to the 5 carbon of the sugar ~~Ribose sugar~~

* Nitrogenous base attached to the 1 carbon of the sugar ~~ribose sugar~~

* Ribose sugar $\rightarrow$ a pentose sugar (deoxyribose in DNA)

(b)

* The 2 strands are held together by hydrogen bonds :-
 * Adenine (A) + Thymine (T)
 = 2 hydrogen bonds.

Cytosine (C) + Guanine (G)
3 hydrogen bonds.

The DNA backbone is made up of alternating sugar and phosphate group.

Q.3(ii)

(a) Two divisions:-

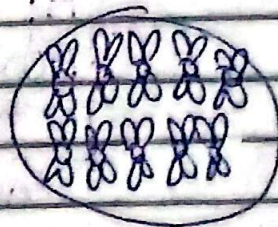
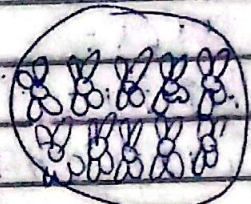
① Meiosis I — the reduction division

② Meiosis II — the second division

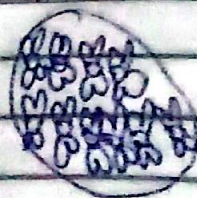
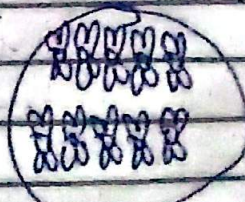
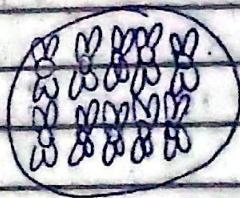
(b) flow chart

Starting cell 20 chromosomes.

↓
meiosis I



meiosis II



c) the cell produced n (haploid)
they will form gamete cells.